FSD2 LAB • EXPERIMENT 10

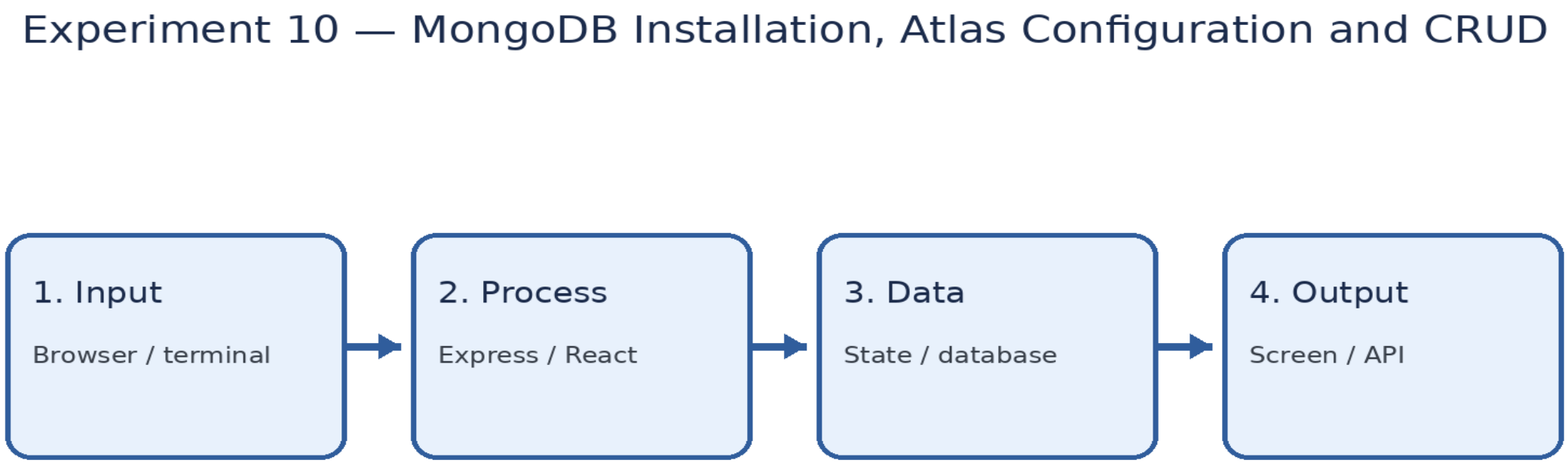
Experiment 10 — MongoDB Installation, Atlas Configuration and CRUD

GitHub-readable preview of the presentation. Download the matching .pptx for the editable PowerPoint.

1. Syllabus tasks

- See the experiment README for the exact syllabus task.

2. Workflow



Workflow: input → processing → data/state → output

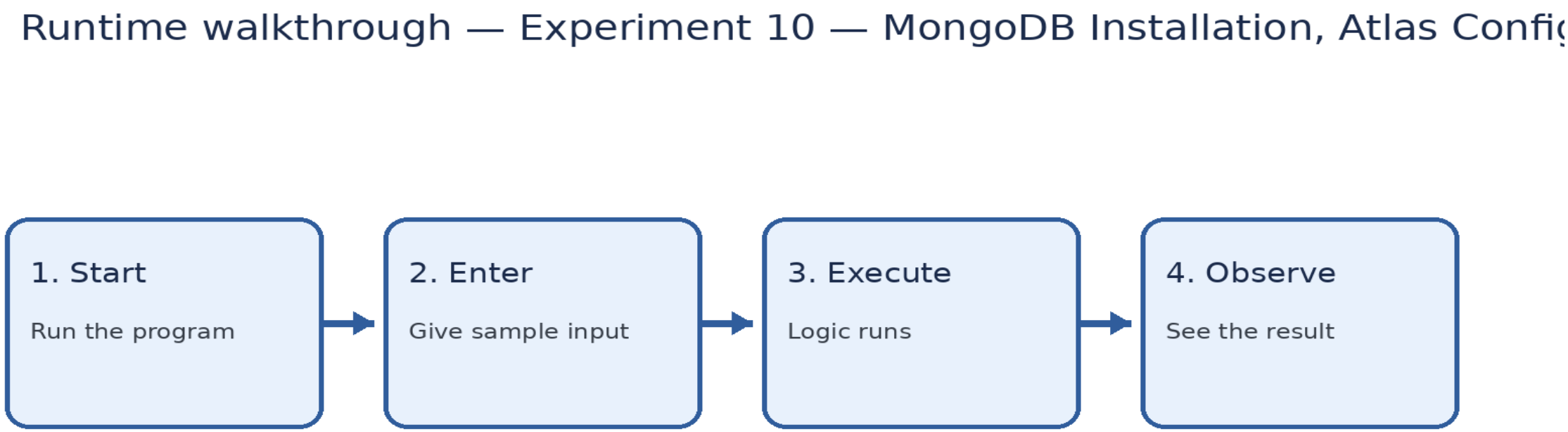
3.1. Code pathway — queries.js

```
/**
 * Experiment 10(b): MongoDB CRUD operations.
 *
 * Run from the repository root:
 * mongosh --file 10-mongodb-installation-configuration-crud/queries.js
 * To use Atlas, pass your connection URI before --file:
 * mongosh "YOUR_ATLAS_URI" --file 10-mongodb-installation-configuration-crud/queries.js
 *
 * This script uses its own fsdLab sample database and prints important results.
 */
```

3.2. Code pathway — queries.js

```
11 db = db.getSiblingDB('fsdLab');
12
13 // Recreate predictable sample state by clearing only this lab collection.
14 db.students.deleteMany({});
15 print('CREATE: insert five sample students');
16 printjson(db.students.insert([
17 { name: 'Anitha', branch: 'CSE', year: 3, cgpa: 8.9 },
18 { name: 'Ravi', branch: 'ECE', year: 2, cgpa: 7.8 },
19 { name: 'Sneha', branch: 'CSE', year: 3, cgpa: 9.1 },
20 { name: 'Kiran', branch: 'MECH', year: 4, cgpa: 6.9 },
```

4. Runtime walkthrough



Runtime: start → enter → execute → observe

5. Run and observe

- Start the server / Vite app or open the MongoDB shell as described in the experiment README.
- Enter the sample input or call the required route.
- Watch the terminal for logs and errors.
- Verify the browser, API response, or database result.

6. Viva questions

- What is the purpose of this experiment?
- Which input is required?
- Which line or command performs the main operation?
- What output should you observe?
- What happens when the input changes?